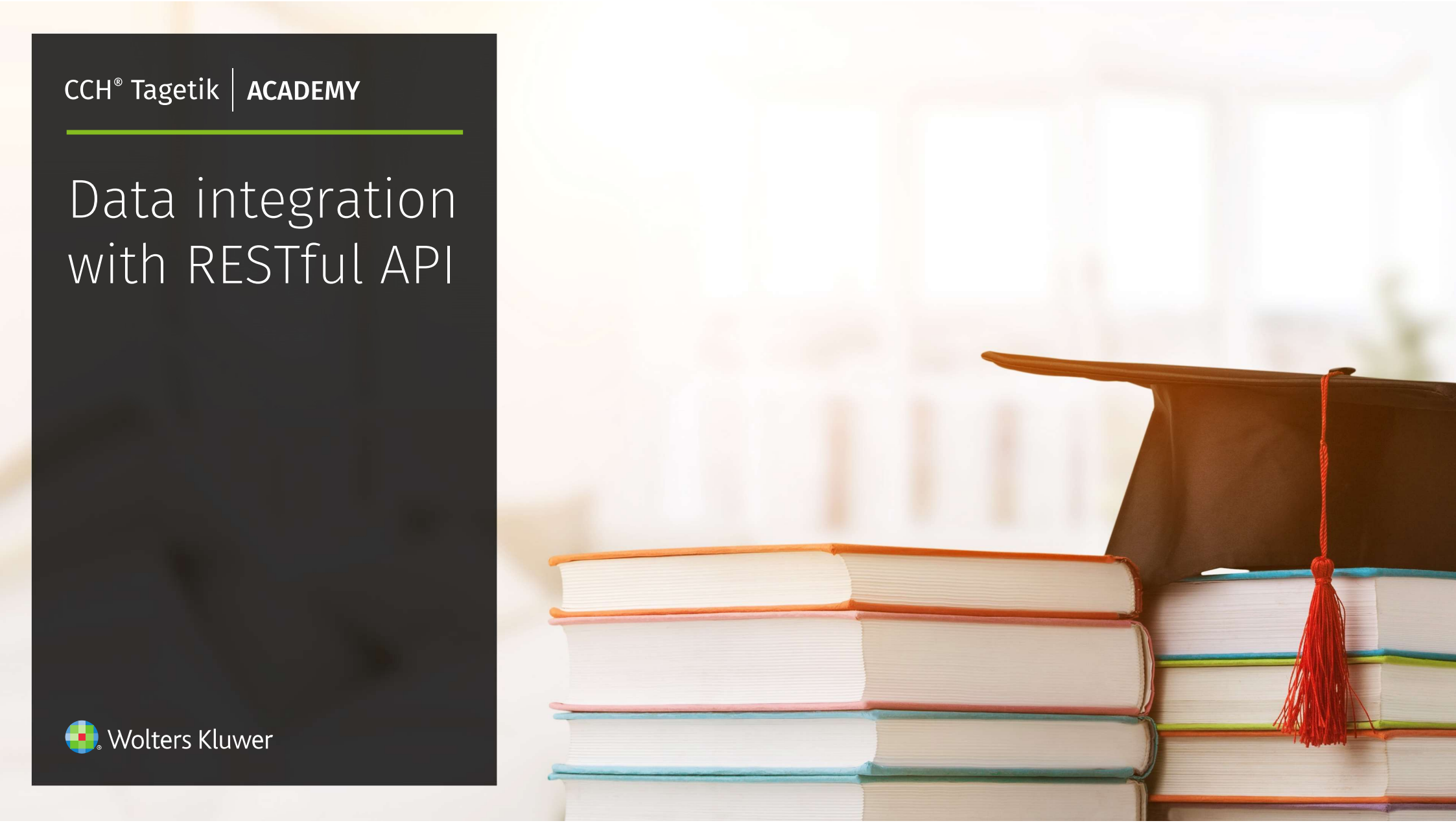
CCH® Tagetik | ACADEMY

Data integration with RESTful API



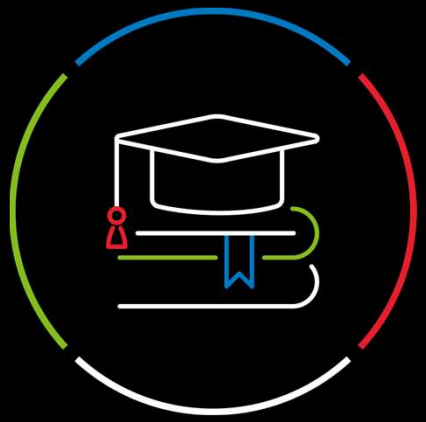
Wolters Kluwer

Change log

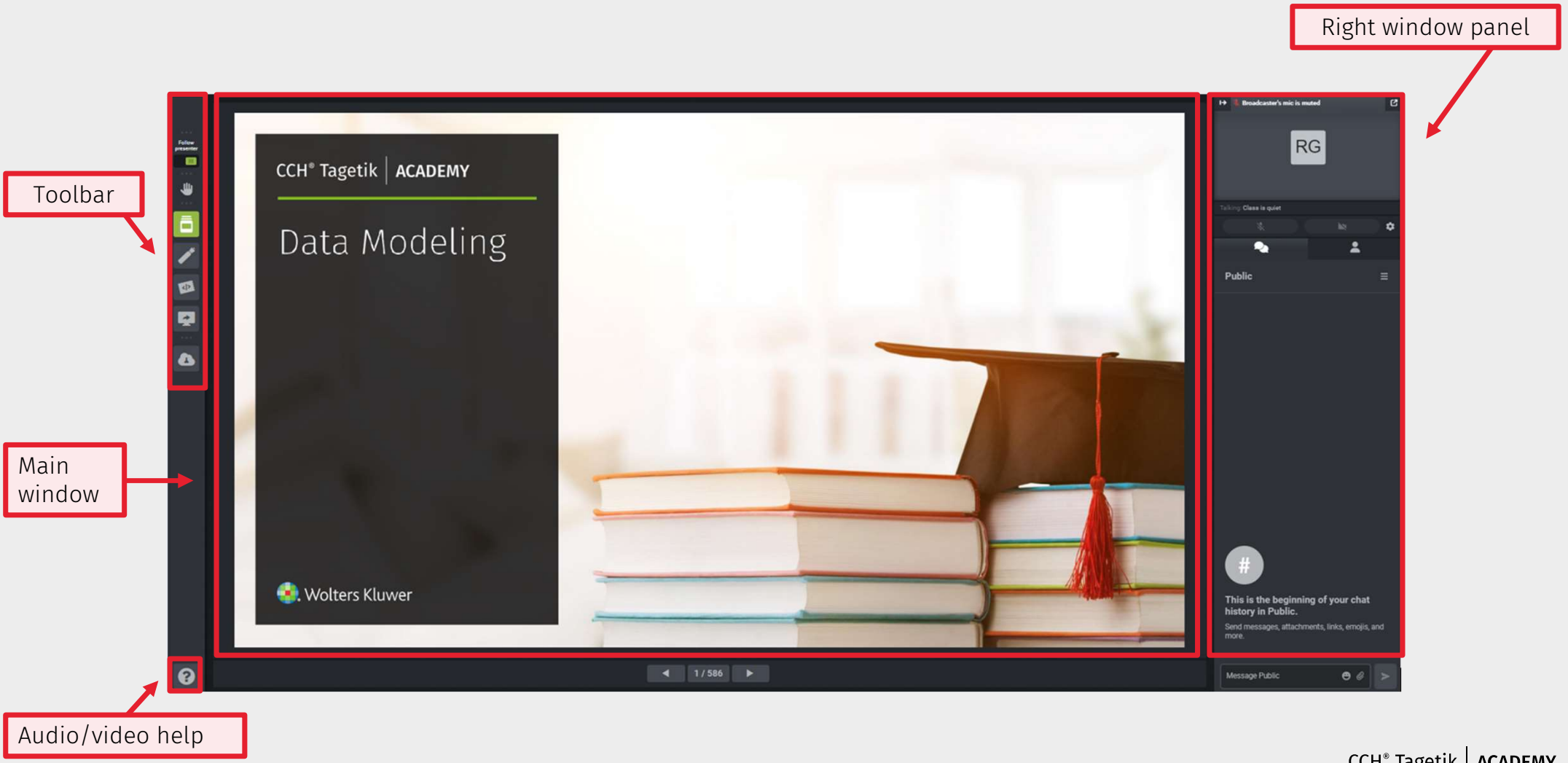
- *Slide URL for Dataset, Virtual Dataset, Datasource*: URL updated
- *Slide AIH data exposed via OData service*: URL updated

Before starting

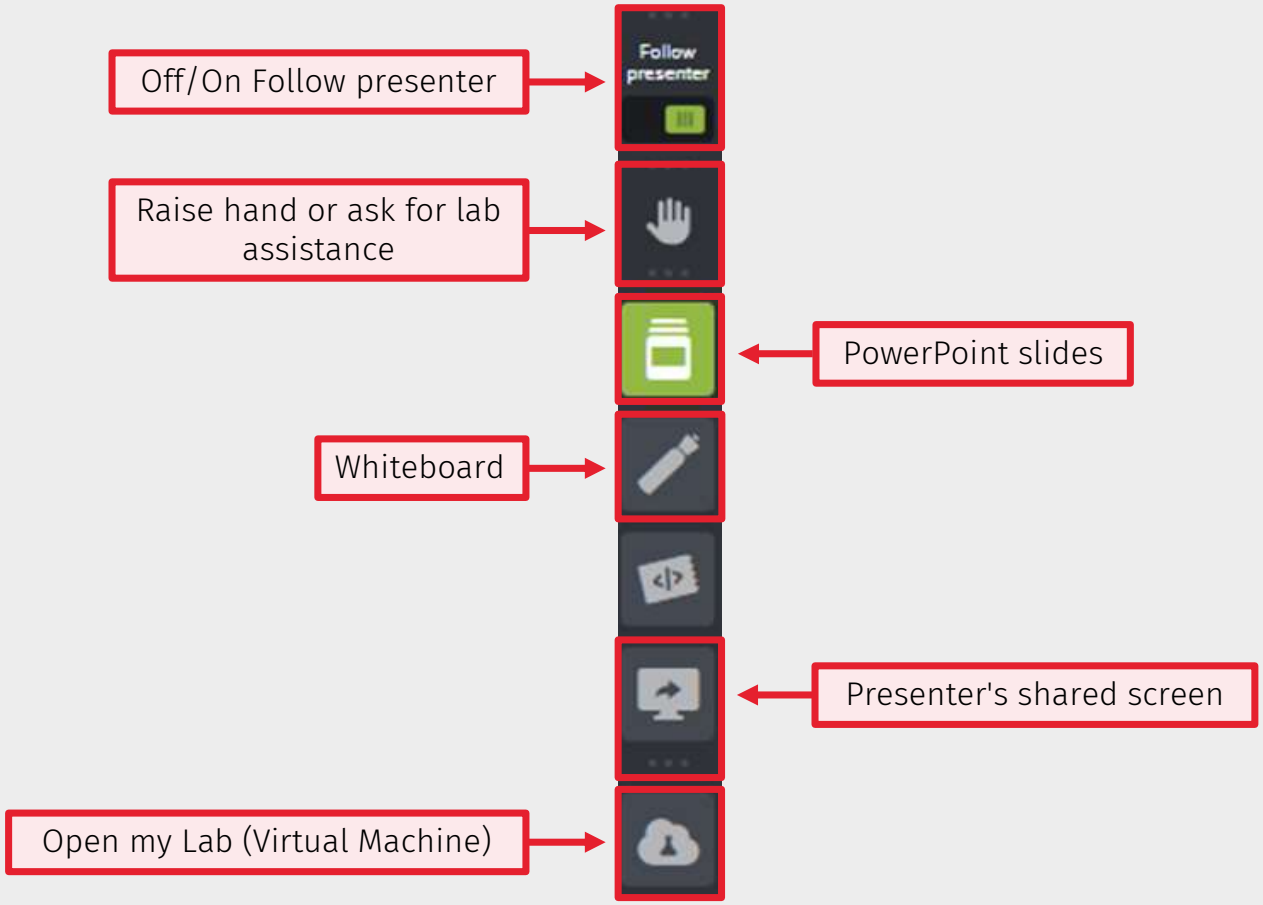
Strigo platform - “How to”



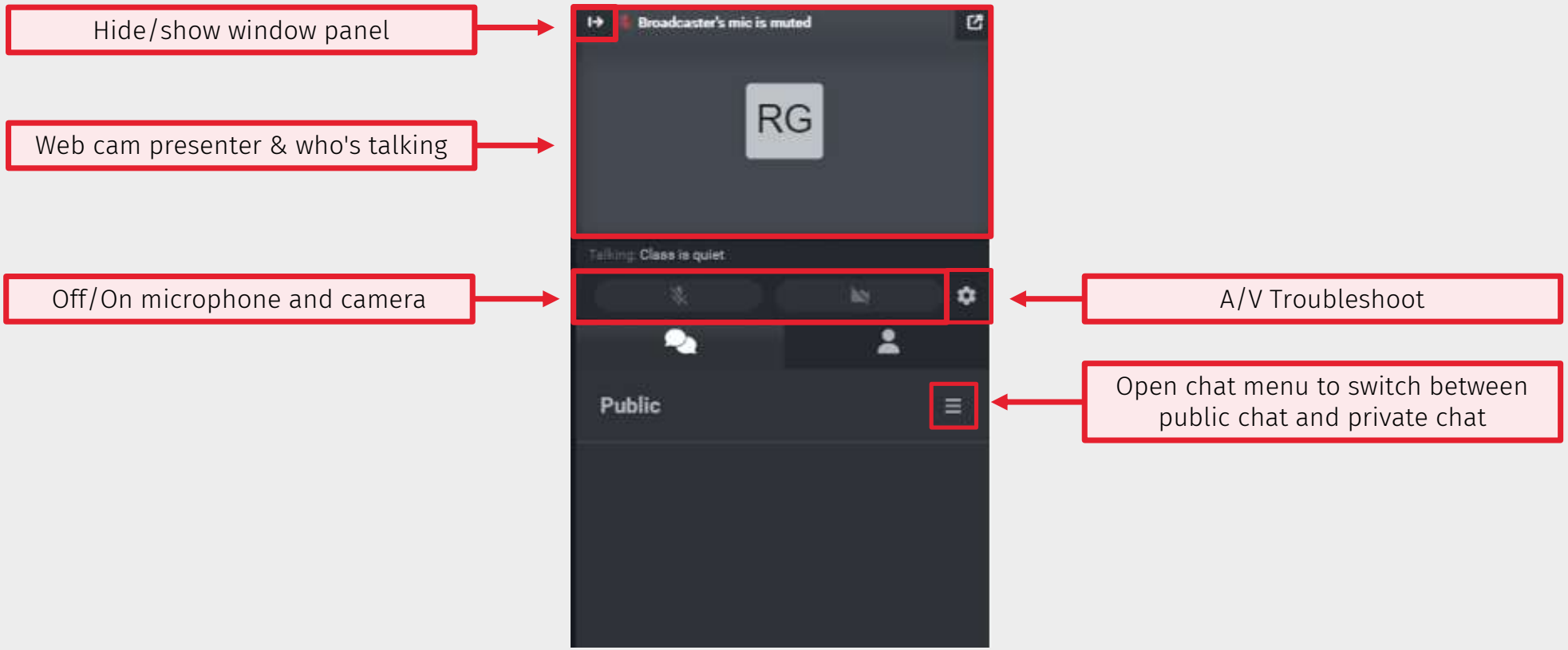
Strigo overview (for Instructor-led classes)



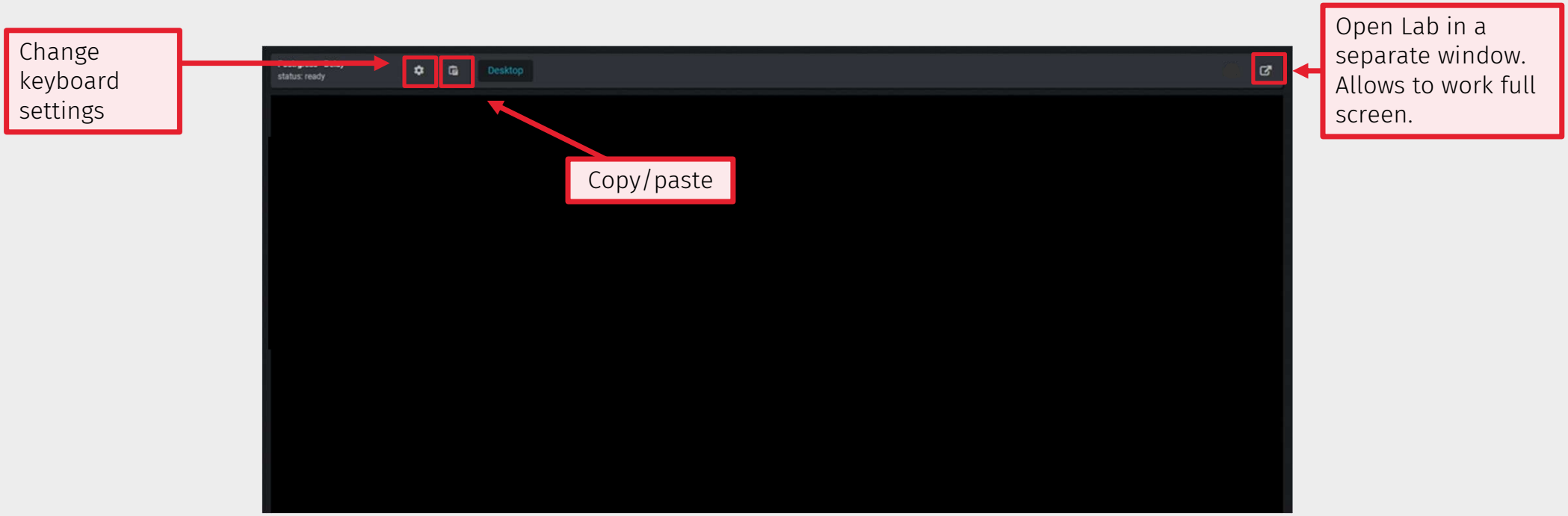
Strigo left toolbar



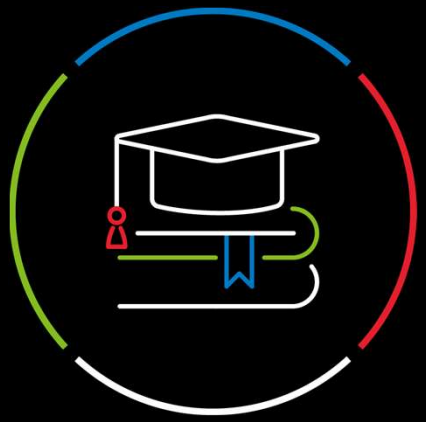
Strigo right window panel



Strigo Lab (Virtual Machine)

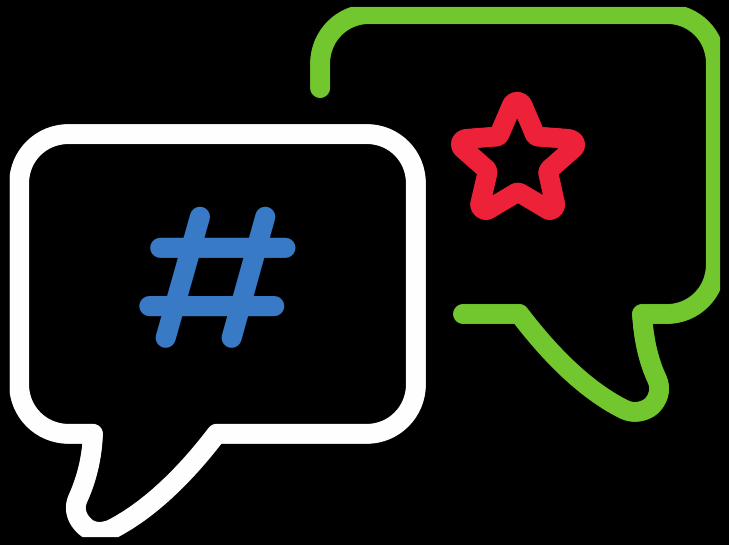


Agenda

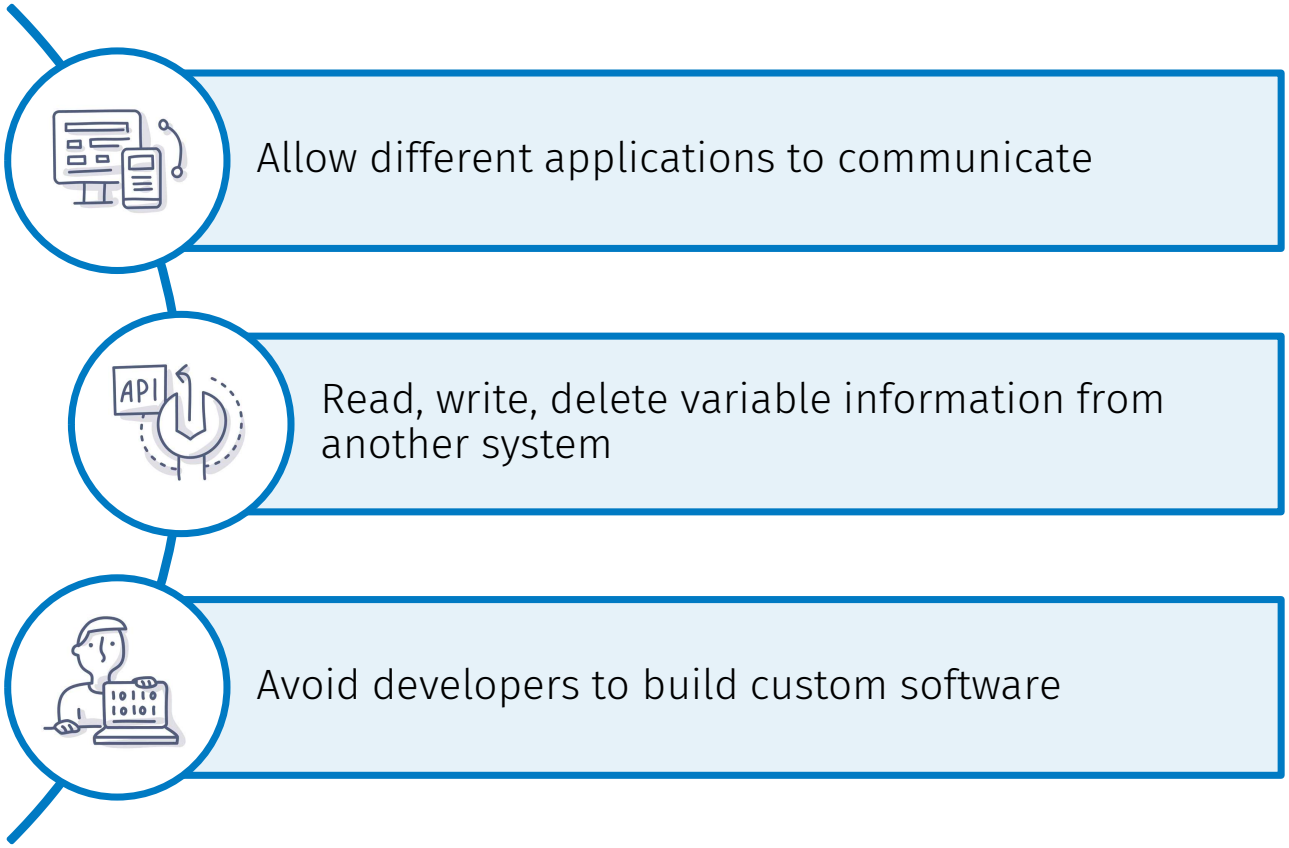


- RESTful API in CCH Tagetik
- Data flow in CCH Tagetik
- How to configure CCH Tagetik to use RESTful API
- Consuming and exposing AW data via OData
- Exposure of Analytical Dimensions
- Data Target
- Audit and OData service
- How to run common IT/Admin operation via RESTful API
- Performances
- Available authentication and OData typical connections

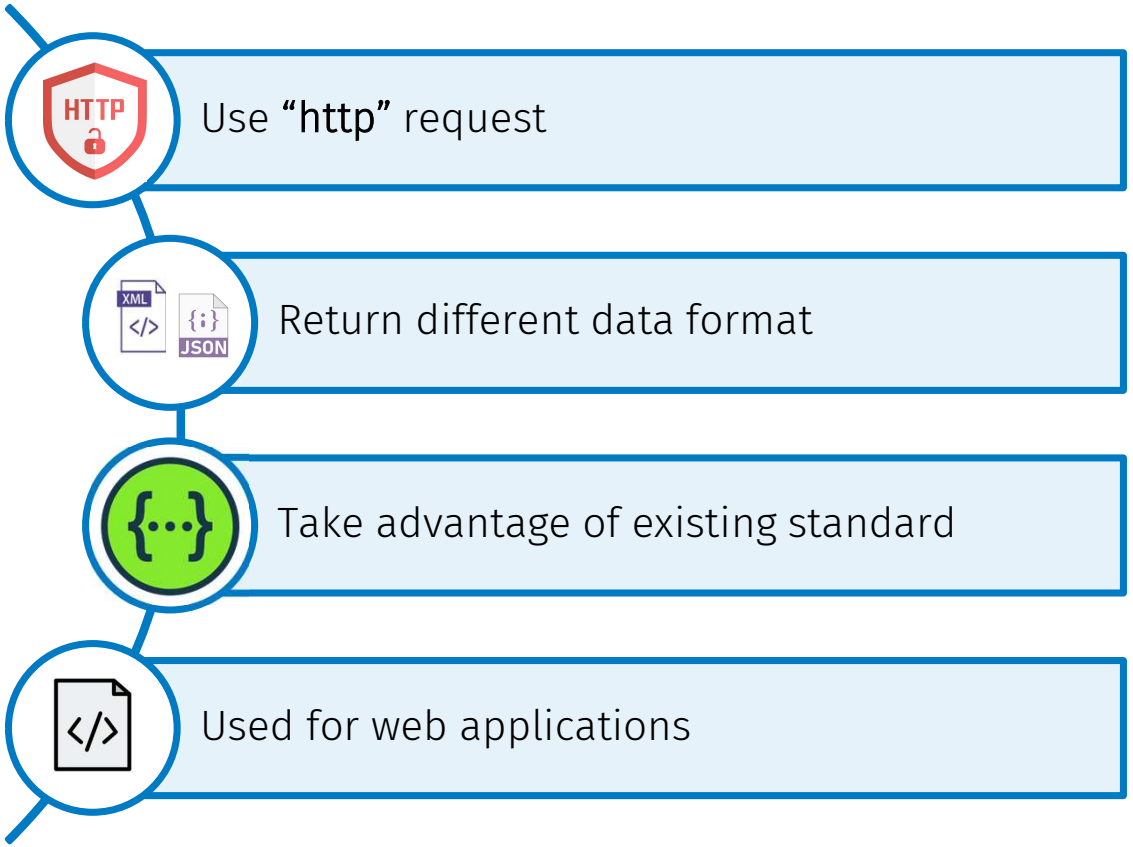
Let’s talk about:
API and RESTful API



What is an API (Application Programming Interface) ?



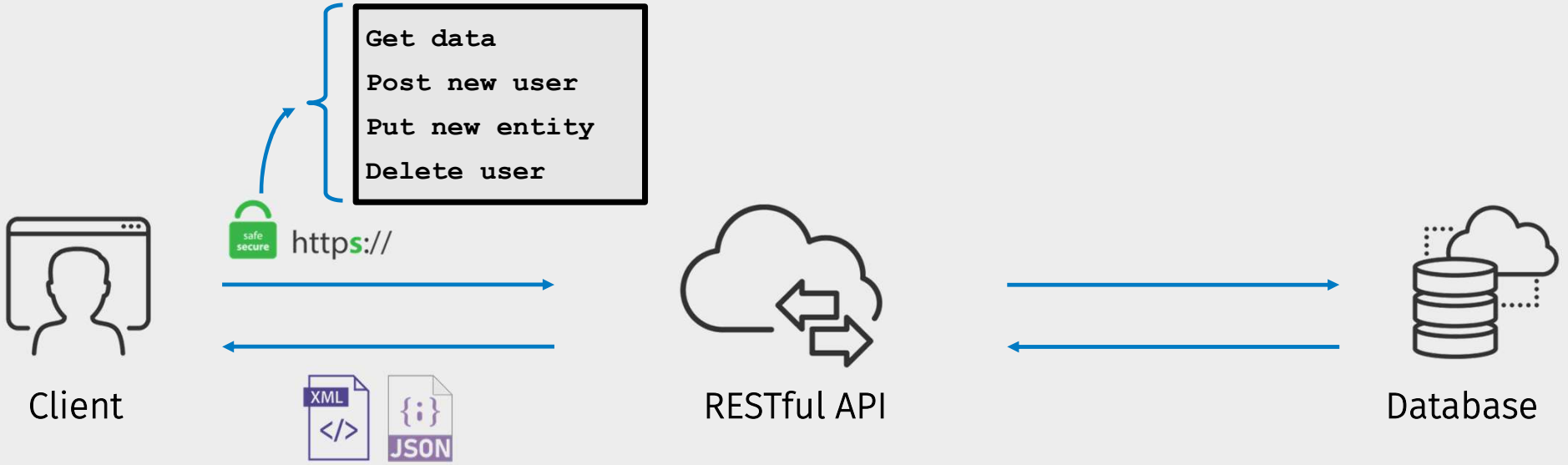
Properties of RESTful API



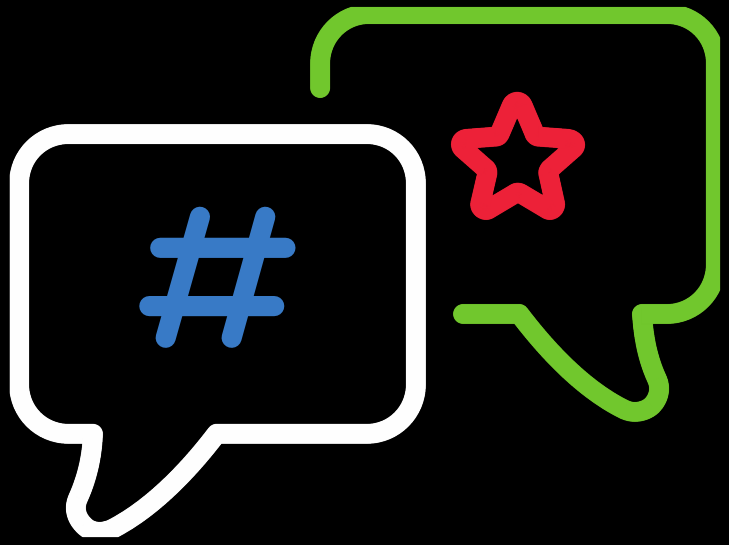
RESTful API Service: How does it work ?

RESTFul API:

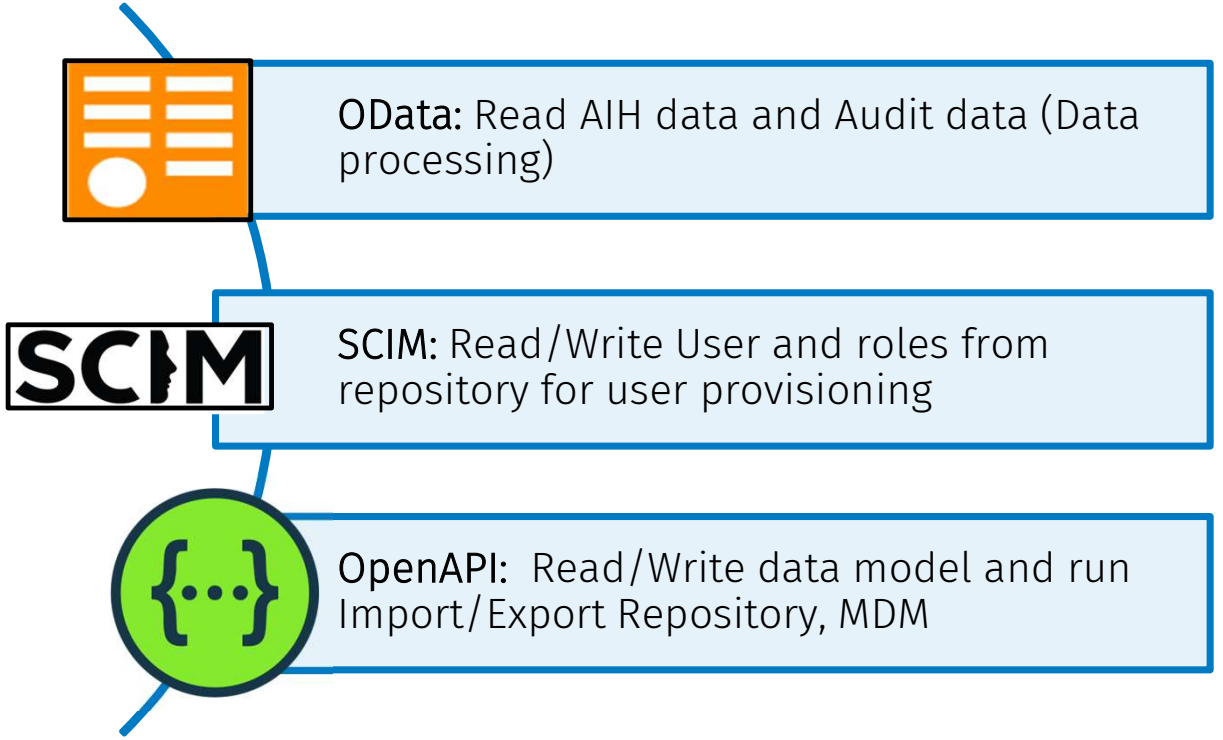
- Receives requests
- Queries the database
- Send back a JSON or XML file



Let’s talk about: CCH Tagetik and RESTful API



Which RESTful API are used from CCH Tagetik ?



When to use RESTfulAPIs in CCH Tagetik ?

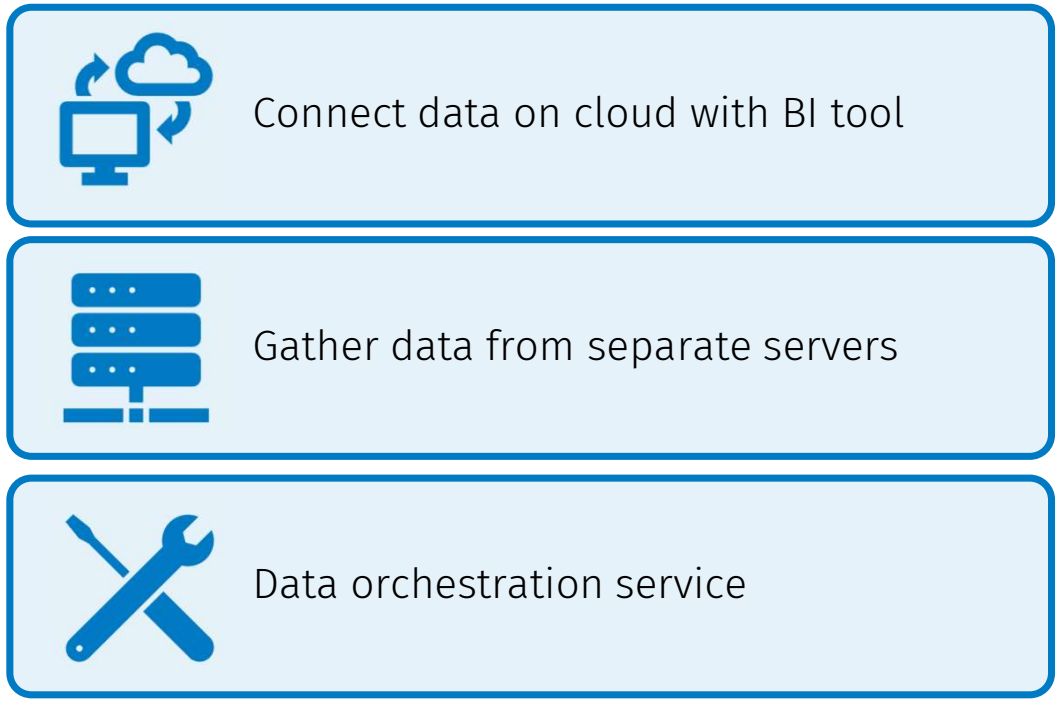
CCH Tagetik application server on cloud

- Difficult to access directly the database

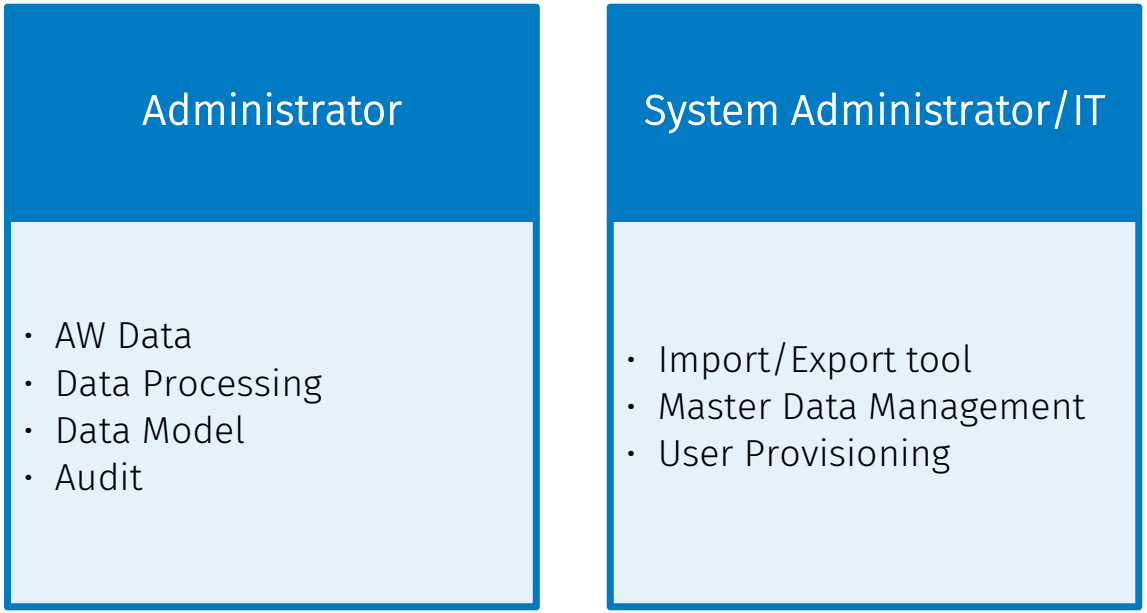
RESTful API

- RESTful APIs eases the access of data directly via HTTPs

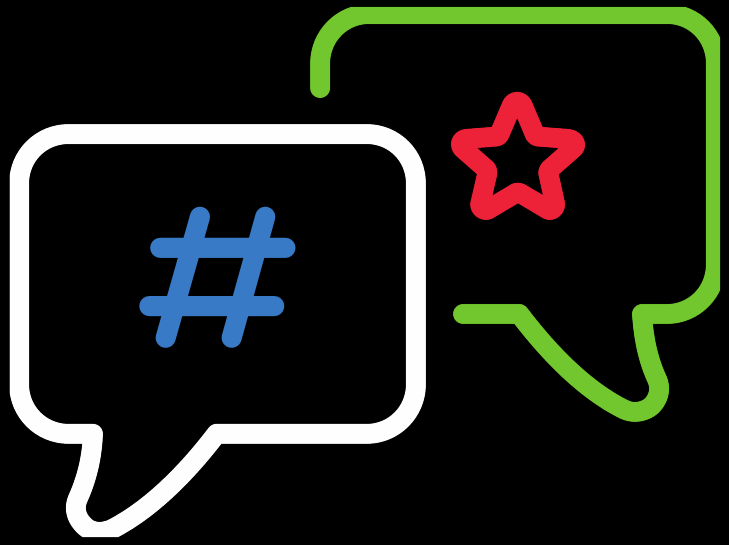
Why to use RESTfulAPIs in CCH Tagetik ?



Who use RESTful API in CCH Tagetik ?



Let’s talk about: OData protocol in CCH Tagetik



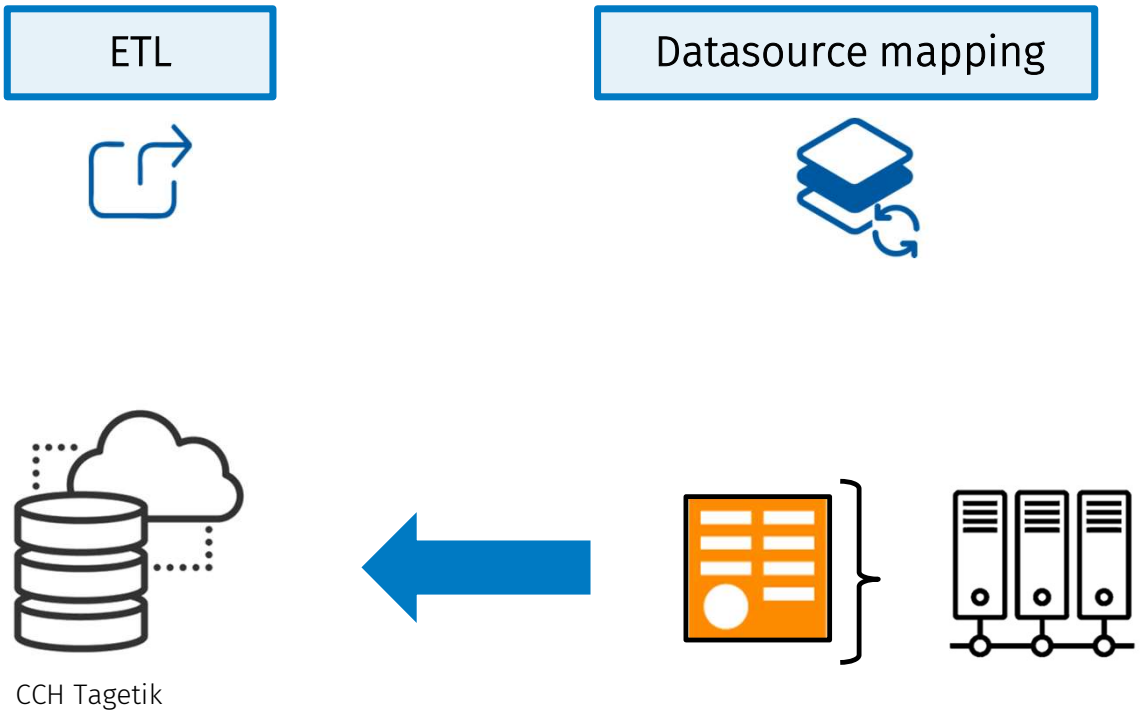
OData service CCH Tagetik

OData (Open Data Protocol) is an ISO/IEC approved, OASIS standard that defines a set of best practices for building and consuming RESTful APIs.

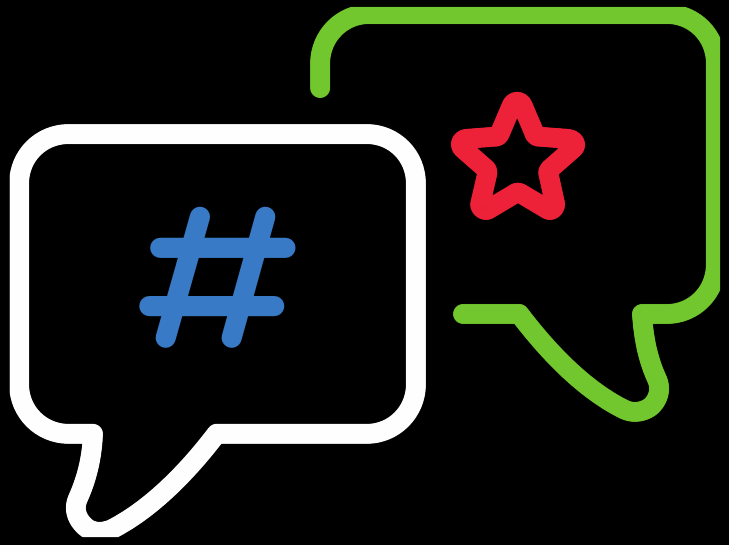
- OData generates a JSON file that the browser translate into XML
- OData service is implemented in different versions
- CCH Tagetik expose AIH data in OData service version 4
- CCH Tagetik can consume OData with version 2 and 4

How does CCH Tagetik consume OData ?

CCH Tagetik can consume OData using:



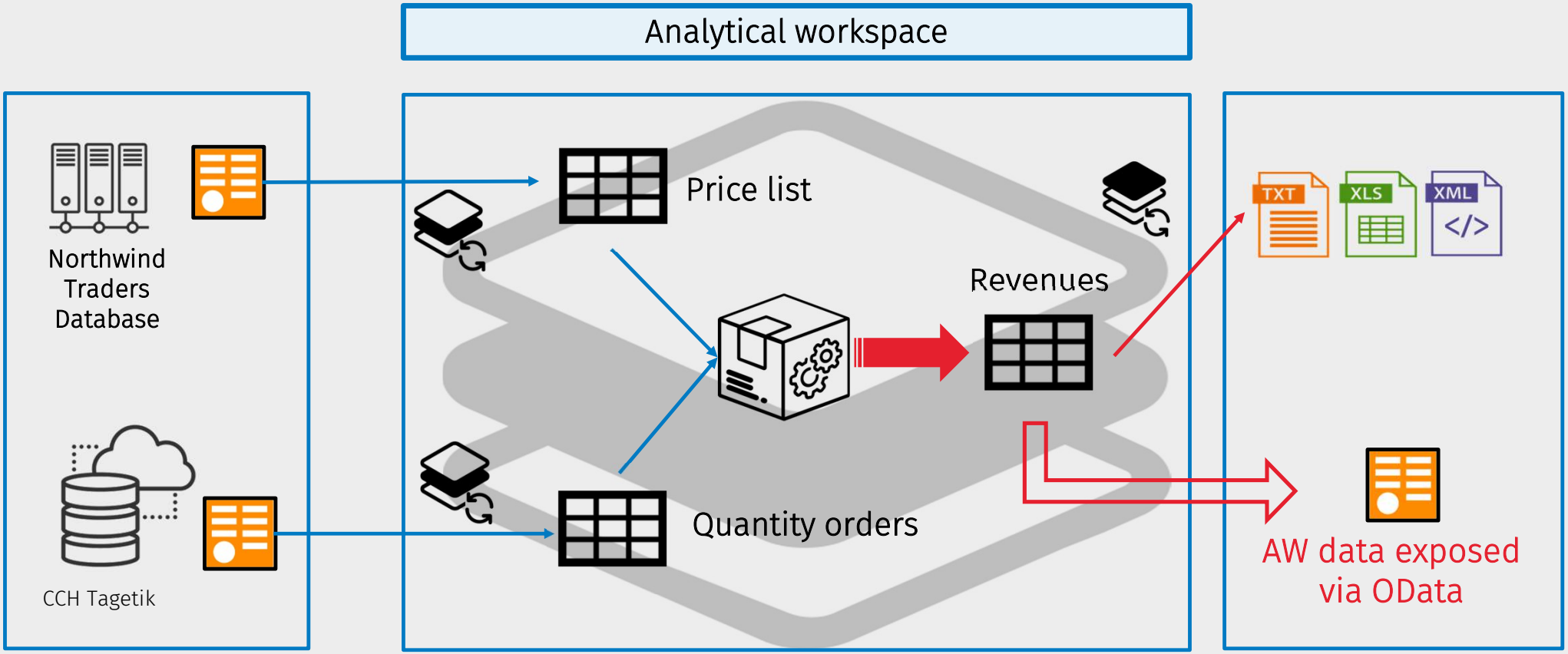
Let’s talk about: Case study: exposing and consuming OData



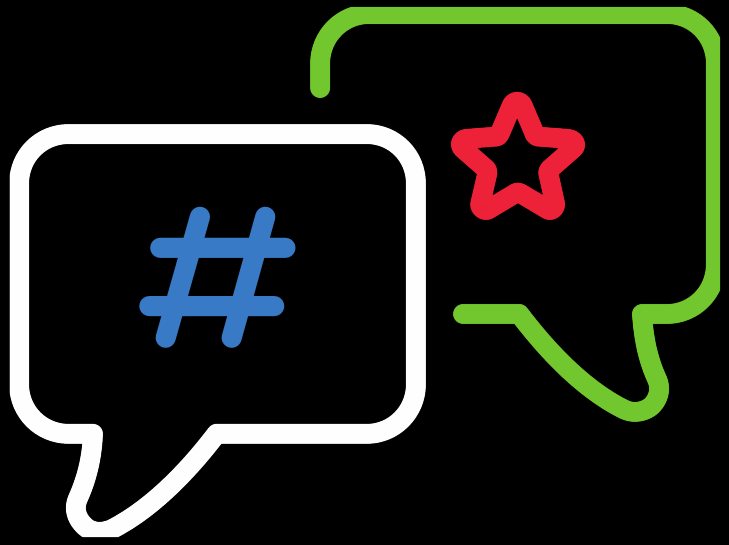
Case study

- Configure CCH Tagetik to use RESTful API
- How to build and consume RESTful API to AW data, dimensions, Audit
- Data Source Mapping with source OData
- Data Target Mapping

Case study: consuming and exposing OData



Let’s talk about: Configure CCH Tagetik to use OData



RESTful API Settings



Enable the use of RESTful APIs from Repository general settings



Enable the function **0004077** to the role of the OData User

Now it’s your turn

Exercise # 1

How to configure the system to use RESTful API



Consuming OData: Endpoint settings

The access to external resources can be set up by defining an **Endpoint** including OData service URL

OData Version

- V2 Returns an xml with specific format
- V4 Returns a json (json more efficent) or xml

Authentication

- Basic (Username and password)
- Oauth2 (Token based)

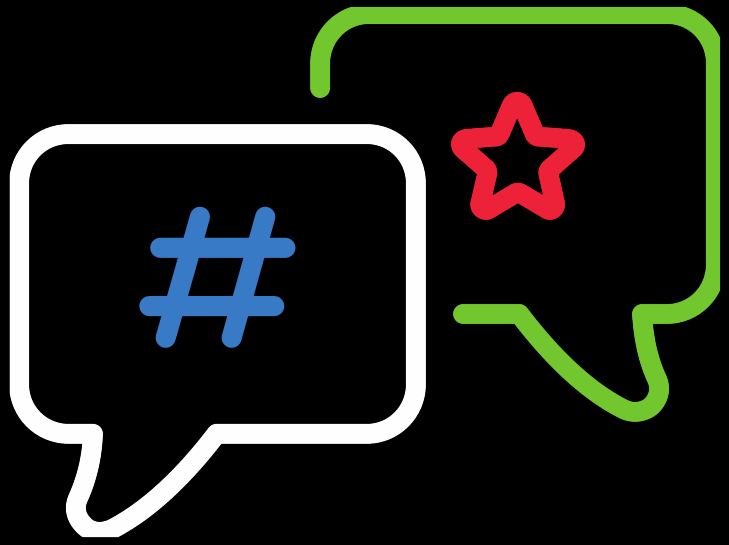
Now it’s your turn

Exercise # 2

Creation of endpoints



Let’s talk about: Consuming OData



Representation of OData resources

OData (Version 4) resources are provided by a service document in JSON (Java Script Object Notation) format, in a list of collections (entitySet)

CCH Tagetik can extract collections called entitySet

- EntitySet holds attributes that can be mapped in Staging Table Fields (ETL) or Dataset fields (AIH)

BI Tool or Excel can be useful to check data in entitySet before setup the mapping

OData service Input Query Options

- All the OData Query options are available to filter data to be read
- Can be specified in Datasource mapping
- Can be concatenated using the clause “&”

Query options list

- **$filter** : https://host/service/Products?$filter=Discontinued eq false
- **$select** : https://host/service/Products?$select=ProductID,UnitPrice
- **$orderby** : https://host/service/Products?$orderby=UnitPrice asc
- **$top** : https://host/service/Products?$top=10
- **$skip** : https://host/service/Products?$skip=10
- **$count** : https://host/service/Products?$count=true
- **$expand** : https://host/service/Customers?$expand=Orders

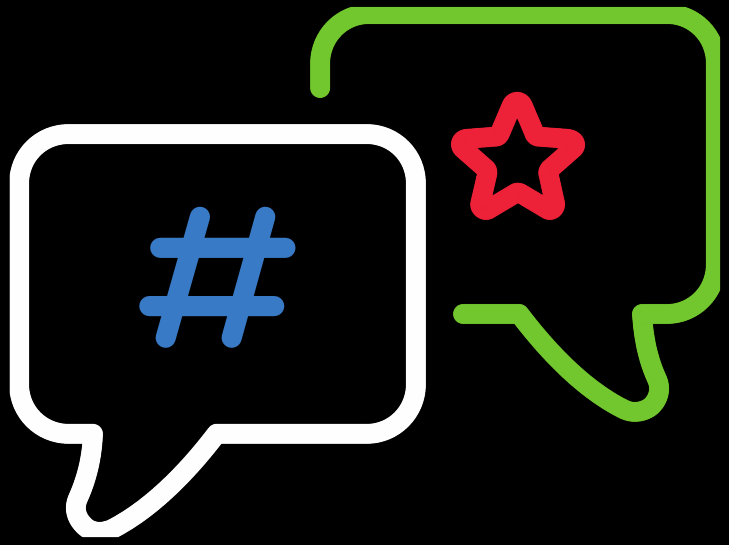
Now it’s your turn

Exercise # 3

Consuming OData



Let’s talk about: RESTful API and Analytical dimensions



Managing Analytical dimensions elements with OpenAPI



OpenAPI

- **Get an element**
- **Create an element**
- **Replace an element**
- **Delete an element**
- **Update an element**
- **Get the list of all elements**

ℹ List of elements

http://*server:port***/tagetikcpm/api/datamodel/v0/analyticaldimension/**database_id/analytical_dimension_label/**elements**

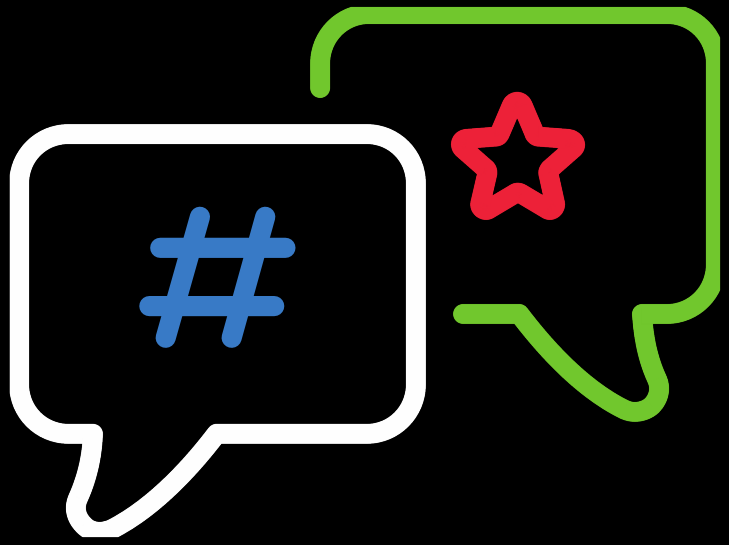
Now it’s your turn

Exercise # 4

How to consume analytical dimensions via RESTful API



Let’s talk about: Expose AIH data via OData service



AIH data exposed via OData service

The URL of the data service for a workspace is:

http://*server:port*/tagetikcpm/api/aih/metadata/v1/*database_id/workspace_code/*Datasets

- Tagetik expose a json file that browser translate into an xml
- Data can be translated in json adding to data service url ?$format=json
- AIH Data are exposed without any limitations (user rights)

The system exposes:

- Physical datasets
- Virtual datasets
- Datasources (Workaround to expose data of FW)

URL for Dataset, Virtual Dataset, Datasource

http://*server:port***/tagetikcpm/api/aih/data/v1/***database_id/***workspace_code/***

Dataset (not partitioned) retrieve data from Reporting Layer

Add ***Dataset_****codDataset****()*** after the URL above

Dataset (not partitioned with filters)

Add ***Dataset_****codDataset(Scenario='2050BDG',Period='12')*

Virtual Dataset

Add ***VirtualDataset_****codVirtualDataset****()***

Datasource

Add ***Datasource_****codDatasource****()***

Dataset Parameters

Parameters are necessary to retrieve data in dataset, virtual dataset or datasource

Possible parameters

- Scenario
- Period
- Entity
- Process

Parameters required depends on

- Partitioning policy
- Dimensions on dataset fields
- Submission type of Process

Dataset Parameters

Parameter's value can be indicated :

List of Parameters

http://*server:port***/tagetikcpm/api/aih/metadata/v1/***database_id/ workspace_code/dataset_code/Parameters**

1) Punctually:

```
Dataset_codDataset(Scenario='2050BDG',Period='12')
```

2) With keyword $ALL that include all values :

```
Dataset_codDataset(Scenario='$ALL',Period='$ALL')
```

OData service Output Query Options

All the OData Query options are applicable to Dataset, Virtual dataset, Datasource

Query options list

- **$select** : https://host/service/Datasource_000003()?$select=Scenario,Amount
- **$orderby** : https://host/service/Datasource_000003()?$orderby=Account asc
- **$top** : https://host/service/Datasource_000003()?$top=20
- **$skip** : https://host/service/Datasource_000003()?$skip=10
- **$count** : https://host/service/Datasource_000003()?$count=true

Now it’s your turn

Exercise # 5

How CCH Tagetik exposes AW via OData



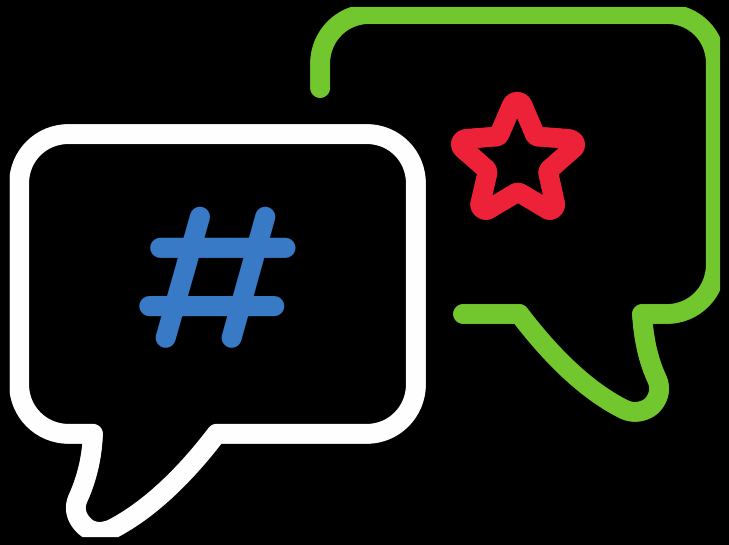
Now it’s your turn

Exercise # 6

Configure datasource mapping when source is OData



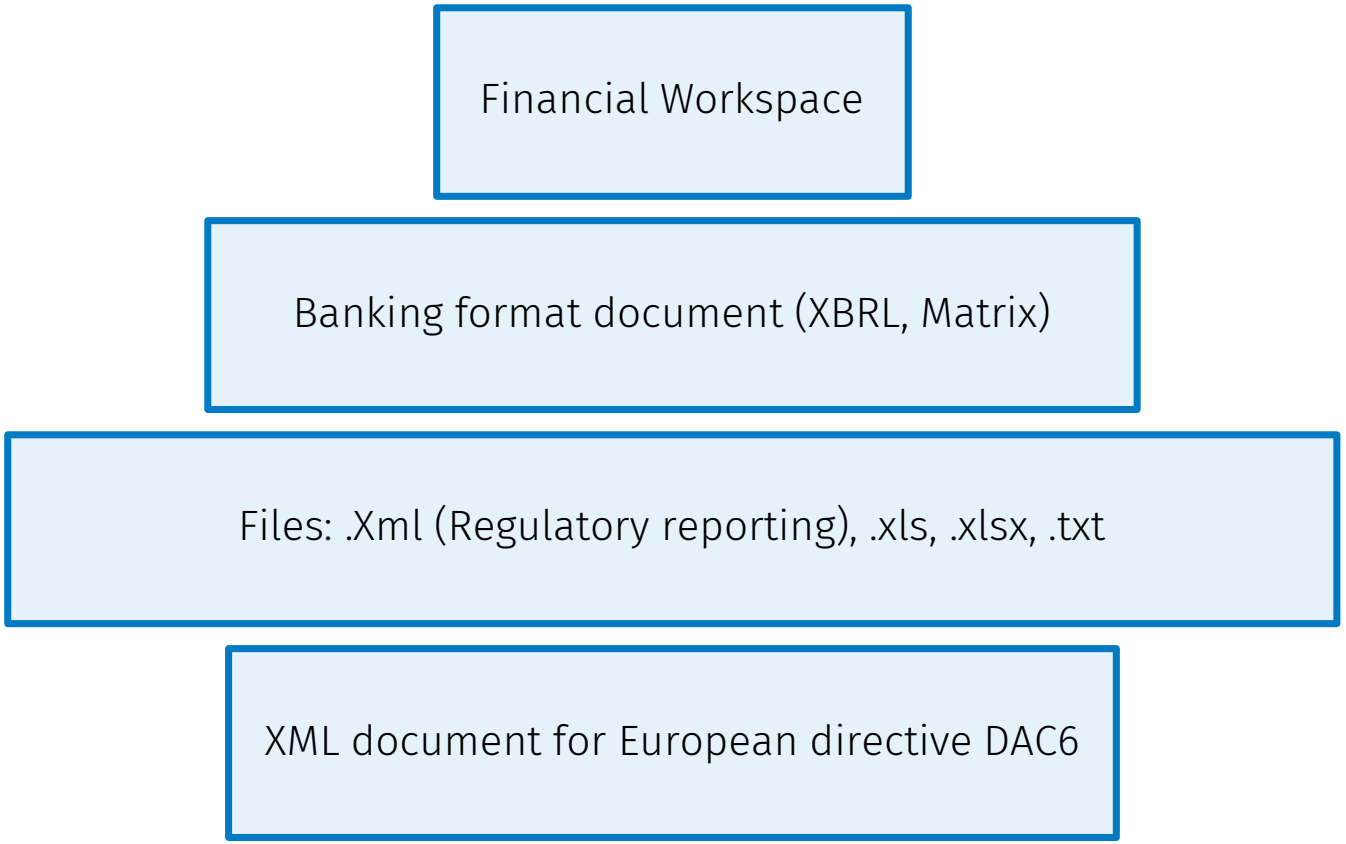
Let’s talk about: Data Target



Data Target mapping

- Typology to export can be chosen while defining the task
- The output files can be input for other systems
- Exported data depends on Process submission and partitioning
- Possible running on nodes

Data Target/Financial workspace mapping



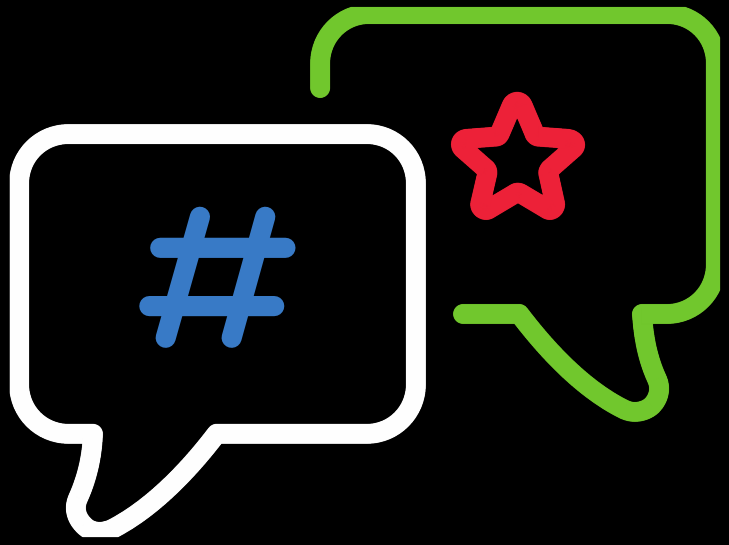
Now it’s your turn

Exercise # 7

Configure data target mapping



Let’s talk about: Data Orchestration and AIH tasks



What is Data Orchestration ?

Is all about making data more useful

- Automate processes and workflows
- Avoid manual processes
- Increase scalability of applications

Helps Sys Admin to manage and automate

- Difficult activities
- Cloud installations
- Complex workflows
- Infrastructure with many servers (provisioning, deployment)

Orchestration tools and Postman



- Tools that lead Data Orchestration
- Orchestration tools can manage and run API



- Postman is an API client for test
- Used (in our case) to run a POST requests
- Run tasks without logging into application
- Simulate use of business workflow

Avalaibility of AIH tasks via API

Data Target mapping : file

Not available for Local download

DTP

Not available for:

- Multidimensional process
- Execution independent from the context

Financial Workspace mapping

Datasource mapping

⚠ Parameters required depends on Process Submission

AIH tasks: Useful URLs

Data Transformation Package

http://*server:port***/tagetikcpm/api/aih/dtp/v1/***database_id/AW_code/DTP_code*

Financial workspace mapping

http://*server:port***/tagetikcpm/api/aih/updatefw/v1/***db_id/AW_code/Dataset_code*

Data Target Mapping:file

http://*server:port***/tagetikcpm/api/aih/dtm/v1/**db_id/AW_code/Dataset_code/DTM_code

Datasource Mapping

http://*server:port***/tagetikcpm/api/aih/loadtoaw/v1/**db_id/AW_code/Dataset_code/typology/DSM_code

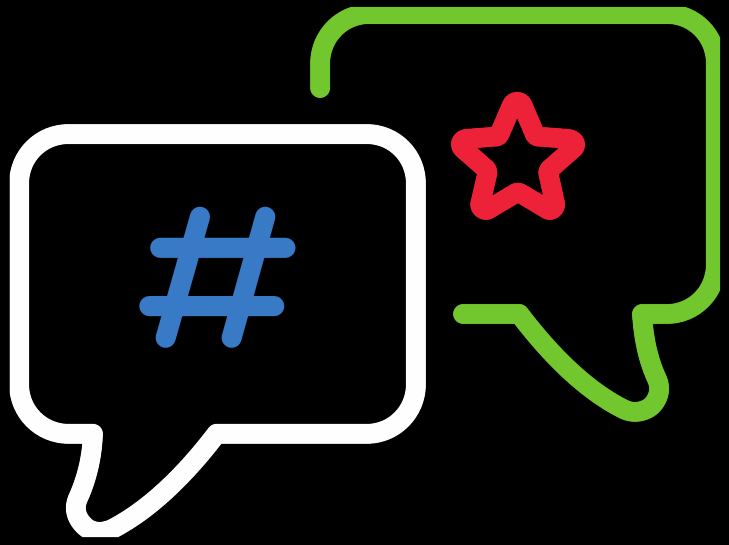
Now it’s your turn

Exercise # 8

Run AW tasks via OData service and check results on exported files



Let’s talk about: Expose CCH Tagetik data model via OData service



Exposing CCH Tagetik Data model via OData service

CCH Tagetik data model cannot be exposed directly via Odata



Datasources can expose indirectly data retrieved from CCH Tagetik data model



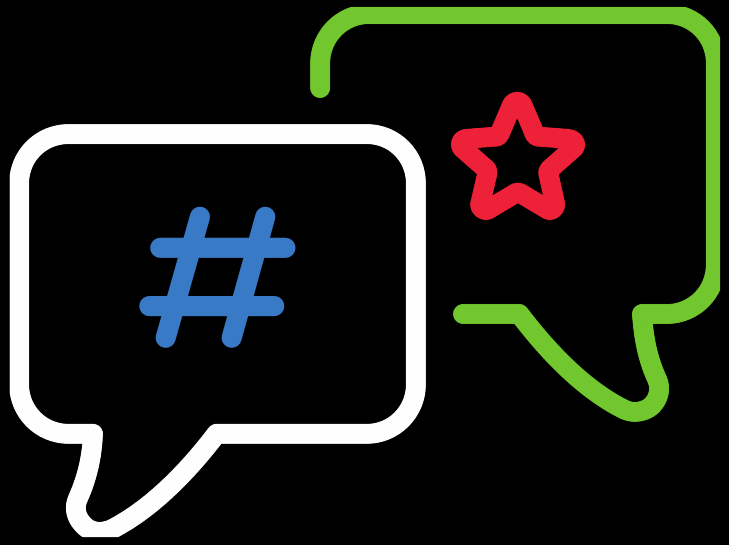
Now it’s your turn

Exercise # 9

Exposure of Dataset, virtual dataset and datasource via OData



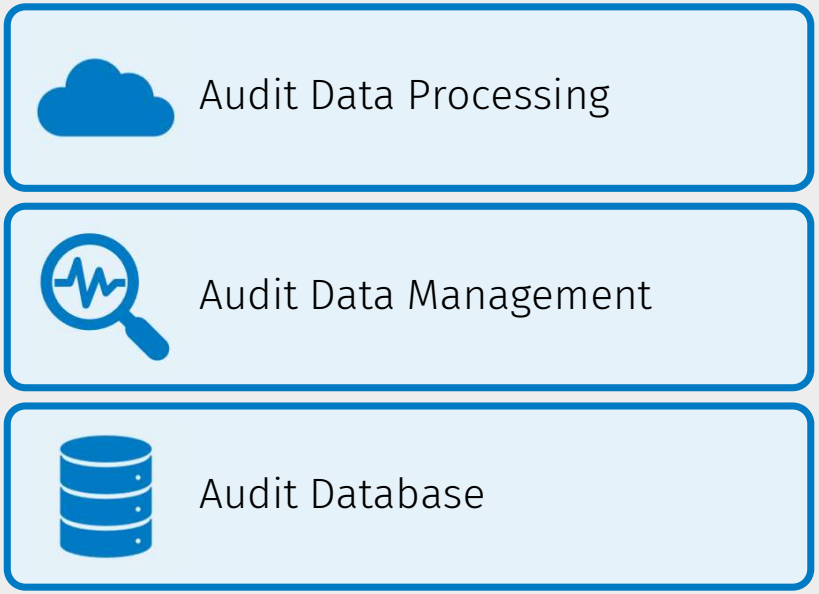
Let’s talk about: Audit and OData



Audit Data via OData service



- Enable role to function **0001286**
- Enable OData



Audit Data: Useful URLs



Audit Data Processing
http://*server:port*/tagetikcpm/api/audit/data/v1/*dataprocessing/*



Audit Data Management
http://*server:port*/tagetikcpm/api/audit/data/v2/*datamanagement/*



Audit Database
http://*server:port*/tagetikcpm/api/audit/data/v1/*database/CONNECT_TO_DATABASE*

Add ***/$metadata*** to access the list of metadata

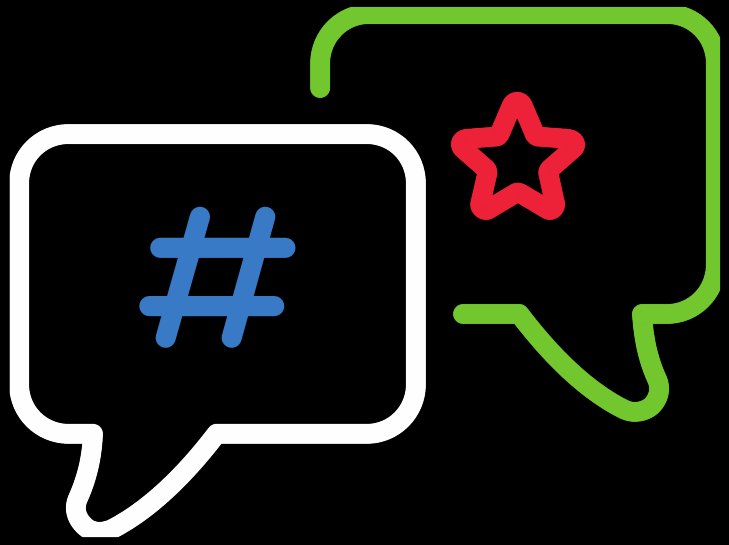
Now it’s your turn

Exercise # 10

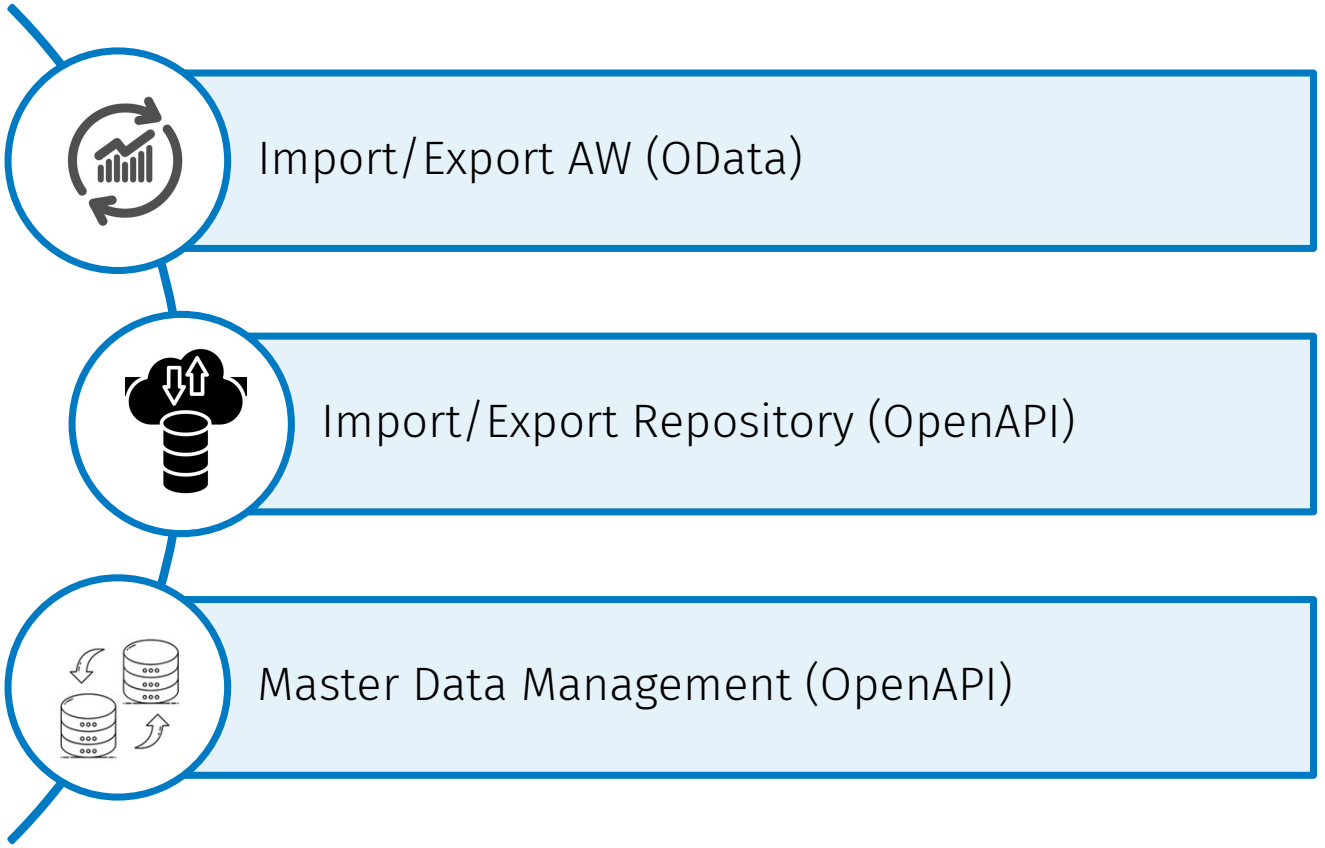
Auditing exposed via OData



Let’s talk about: Running Data Processing via RESTful API



CCH Tagetik Data Processing run via RESTful API



Import Export AIH: URL and parameters

Export AIH URL

http://*server:port***/tagetikcpm/api/aih/export/v1/***database id/workspace id*

Parameter

{"data":"true/false"}	Allows to export or not Data of AW

Import AIH URL

http://server:port**/tagetikcpm/api/aih/import/v1/**database id/workspace id

Parameters

• ***{"filenamefile":""}***	Import this file from *writable folder*
• ***{"dir":""}***	Read the file from "dir" folder
• ***{"endpoint":""}***	Endpoint address if file is on Endpoint

Export Repository: URL and parameters

Export Repository URL

http://*server:port***/tagetikcpm/api/admintools/export/v1/***database_id*

Parameter

{"zipMaxSize":"2048"}	Maximum size of file produced (default 2048 MB)

Import Repository: URL and parameters

Import Repository URL

http://*server:port***/tagetikcpm/api/admintools/import/v1/***database_id/**

Parameters

- ***{"filename":""}*** Import this file from writable folder
- ***{"dir":""}*** Read the file from "dir" folder
- ***{"endpoint":""}*** Endpoint address if file is on Endpoint
- ***{"emptyDb":""}***
- ***{"deletePartitions":""}***

Now it’s your turn

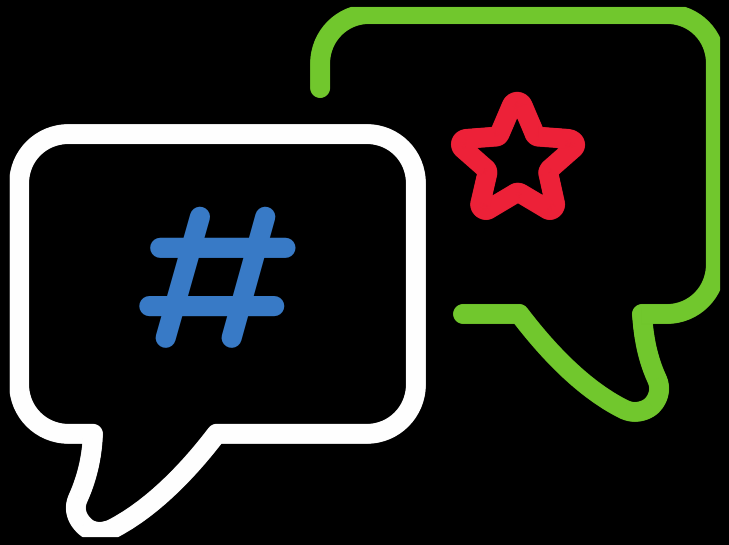
Exercise # 11

AW Import/Export, MDM and Export repository via RESTful API



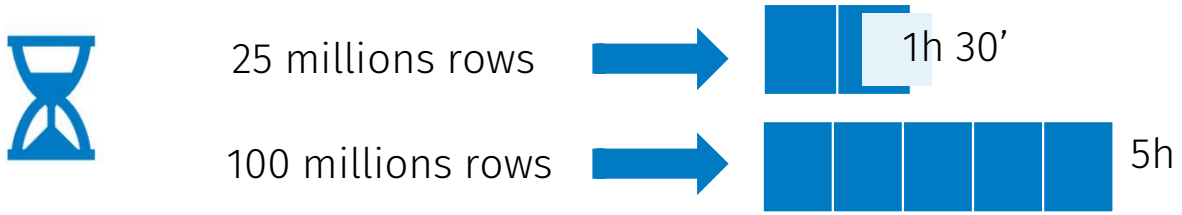
	OData Service	OpenAPI	SCIM
AIH Data	X		
AIH Metadata	X		
Auditing	X		
AIH Data processing	X		
Data model hierarchies	X		
Data model		X	
MDM		X	
Repository Import/Export		X	
User Provisioning			X

Let’s talk about: OData performance



OData service performance

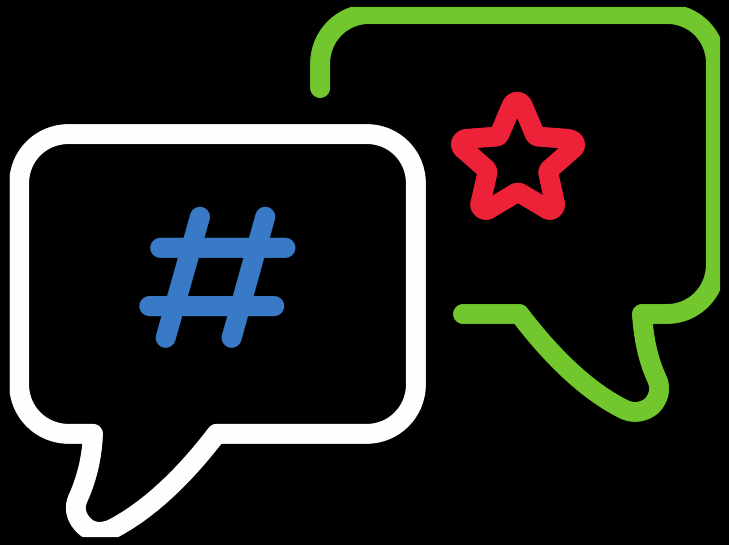
Test performed in reading phase on a dataset with 25 columns



Tested with data packet size (Chunk): 250.000 rows

- Time increase is proportional to number of (rows X columns)
- Number of rows (Chunk) can be set as option in System Resources
- Chunk depends also on available memory

Let’s talk about: Authentication and OData connections



Basic Authentication



Authentication via Local Tagetik User assigned to a role with OData function enabled

Username and Password are the same of Local Tagetik User



- Easy to setup
- Minimal Cloud team intervention



- Single Sign On not supported
- Weak level of security

Basic authentication connection: What to do?



Cloud team has to whitelists IP that pull requests to CCH Tagetik



Local user with proper role and attributes has to be created



- PowerBI allows only Basic Authentication
- Power BI with Auth2 Authentication require a connector that CCH Tagetik doesn't develop

OAuth2 Authentication

Preferred solution when Single Sign On is required

Authentication token based, with reference users created in CCH Tagetik for each different scope

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- Level of security Stronger than Basic
- Single Sign On implementation supported

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- Cloud team and local IT setup required and more complex (next releases it is going to be easier)

OAuth 2 Authentication: What to do?



Cloud intervention to whitelist IP, setup configuration files and manage authentication

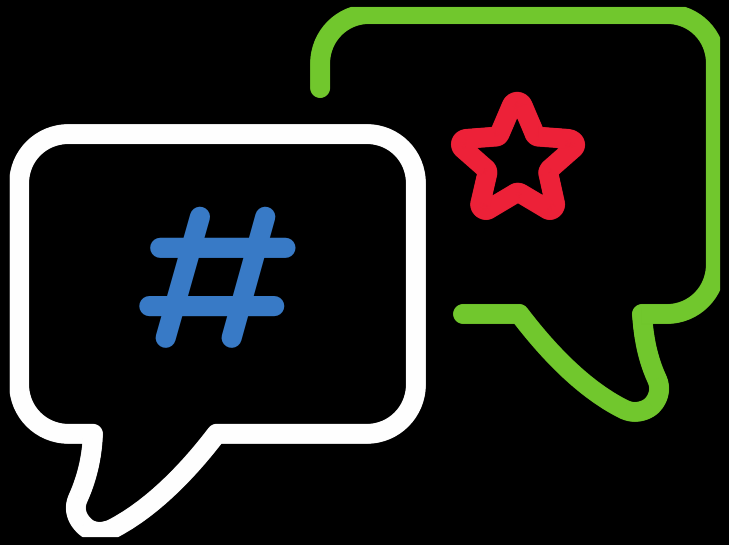


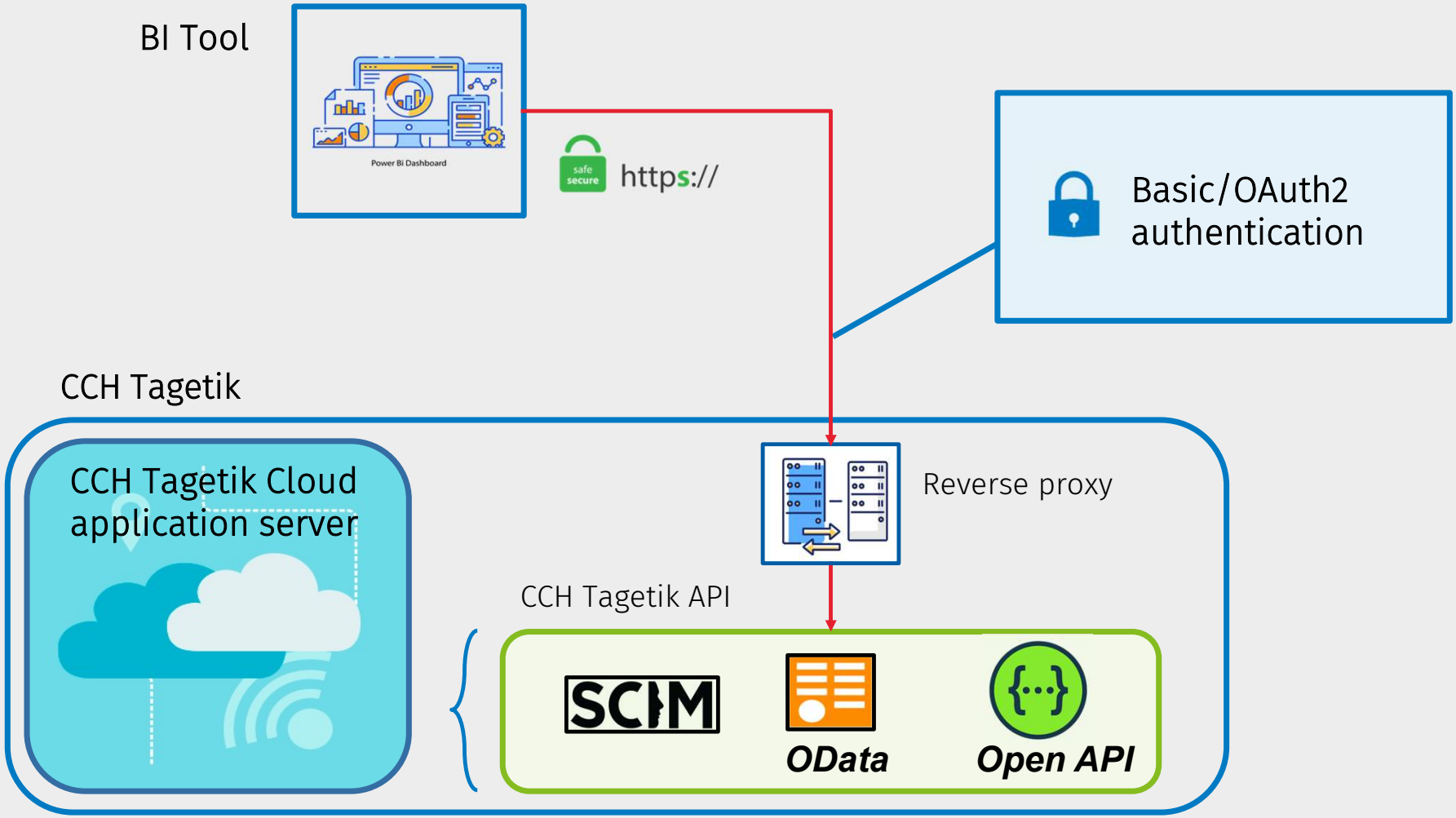
Create 1 reference user for each different scope in the Client IP. These users will not access directly to the application

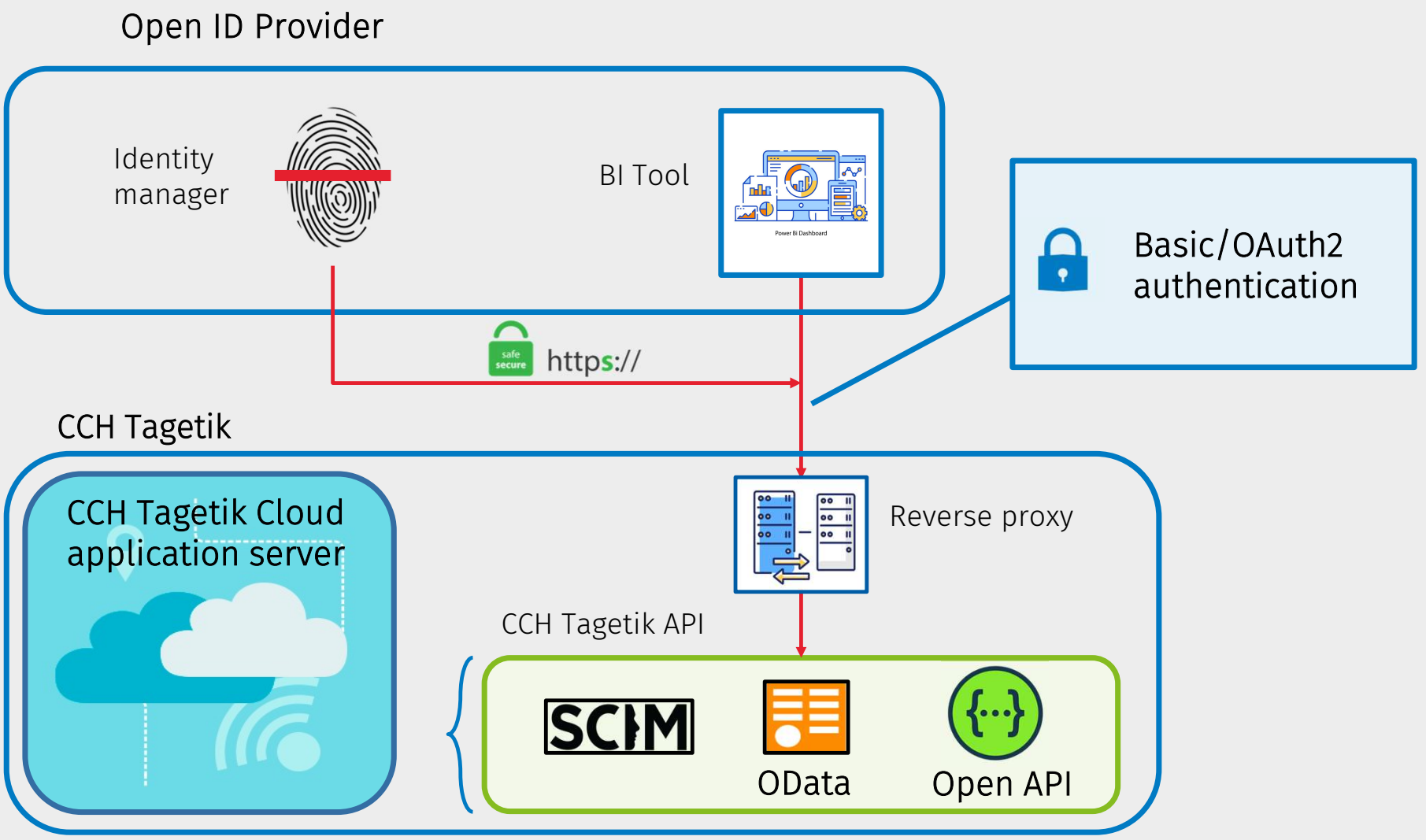


OAuth2 authentication is token based. Requires IT intervention to create the token involved in authentication

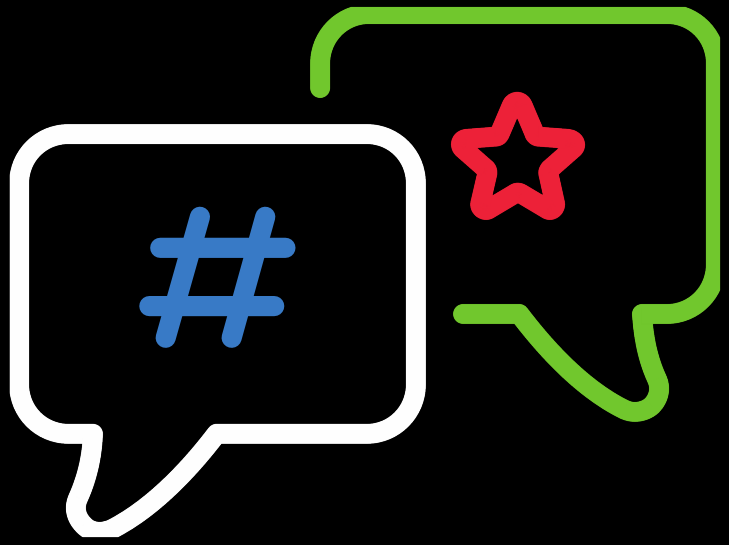
Let’s talk about: Example of installation framework




BI Tool
Power Bi Dashboard
safe secure
https://
Basic/OAuth2 authentication
CCH Tagetik
CCH Tagetik Cloud application server
Reverse proxy
CCH Tagetik API
SCIM
OData
Open API


Open ID Provider
Identity manager
BI Tool
Power BI Dashboard
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Open API

Let’s talk about:
Q&A





Takeaway

Join CCH Tagetik Insiders Hub!



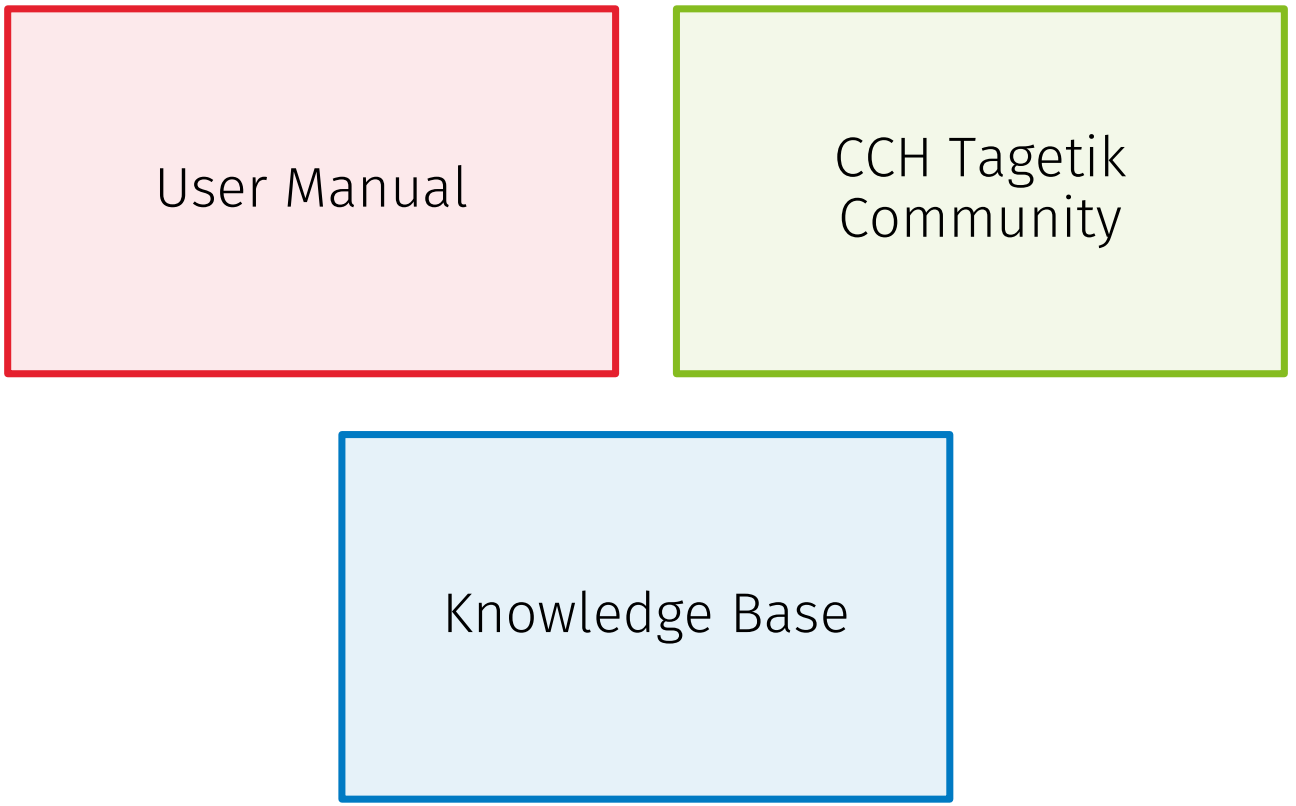
Here's the way the hub works

1. Log into the CCH Tagetik Community and click on the Insiders tab
2. Poke around and complete challenges earning points to redeem rewards
3. Get recognized for all the support you give CCH Tagetik

It's that easy!

Join CCH Tagetik Insiders now!

What other learning assets are available?



What's next?

Visit the Academy webpage

Next training and/or Assessment

Mail to info.academy@wolterskluwer.com


Thank you
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